

The Shape of Seeing the City

Propose Review | Minsup Chung

Collective Understanding

Objective of Project

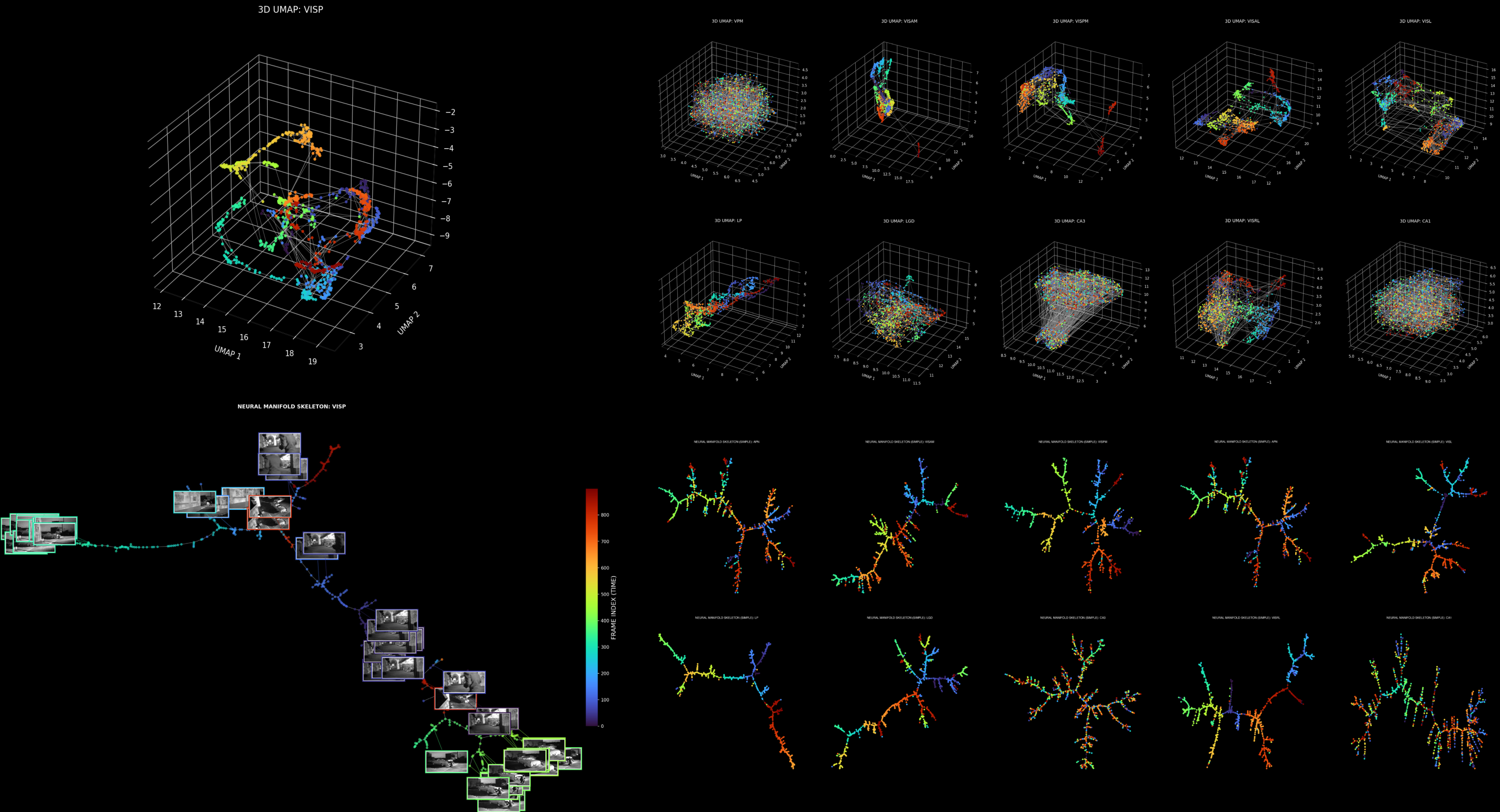
This project reveals how urban scenes unfold within high-dimensional neural geometry and aims to make collaborative system that general public could involved in inducing knowledge..

Using spike data recorded while a mouse watches city videos, we transform population activity with PCA, UMAP, and persistent homology to uncover region-specific manifolds that reorganize visual complexity into distinct shapes.

By turning these manifolds into a collectively explorable space, the project invites non-experts to map, interpret, and engage with neural structures that are typically reserved for scientific analysis.

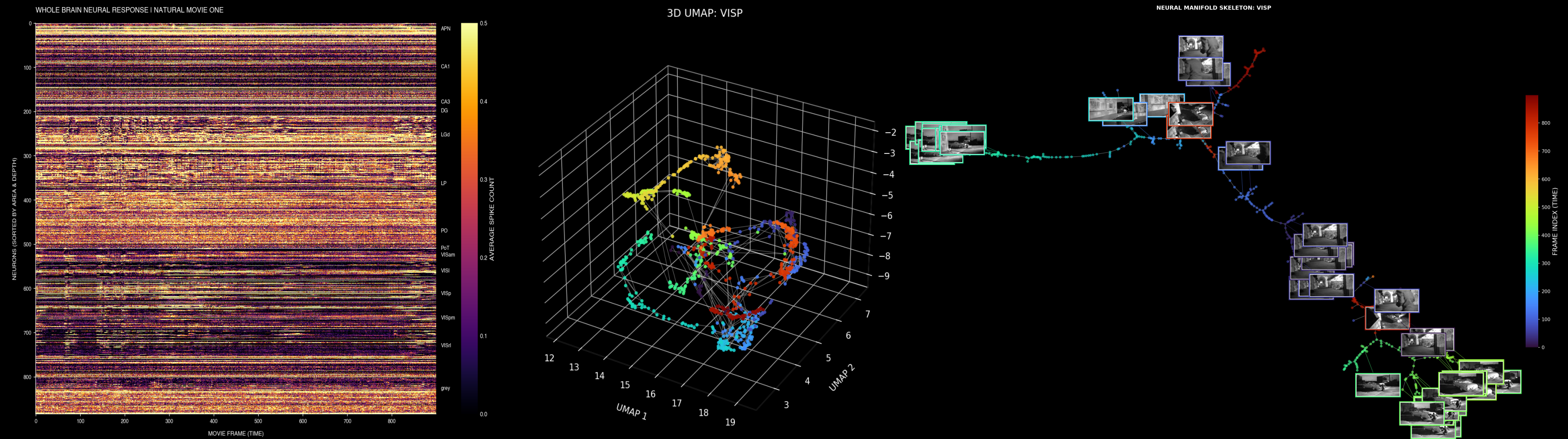
Output

Collective Mapping of Neural Manifold - How Brain Encodes the World?



Output

From Raw Neural Spiking Data to Geometrical Representation



Raw Neural Spike Data



Low Dimensional Representation



Neural Manifold

What is Explainability?

Level of Understanding

Ability to Understand the System's Behavior

The capacity to clearly describe why a system produced a certain output, action, or pattern.

Transparency of Internal Logic

Making the internal rules, mechanisms, or structure of the system interpretable to humans.

Human-Readable Reasoning

Providing an explanation in a form that humans can comprehend—concepts, narratives, diagrams, rules, etc.

Simplicity Without Oversimplification

Giving explanations that reduce complexity while preserving essential truths.

Core problem in Understanding Neural Network

Understanding Through Topological Structure of Network

High-Dimensional Data Complexity

Neural activity and connectivity exist in extremely high-dimensional space (3D + time + function), making meaningful pattern extraction difficult.

Correlation vs. Interpretation Problem

Even when correlations in brain activity are found, it's hard to understand the meaning or function behind them.

No Ground Truth

The brain contains both metric-based navigation systems (grid-like) and memory-based systems (place-like), and we lack a combined model that integrates them.

Data Structure of Neural Spike

RAW Information of Neural Spike Data

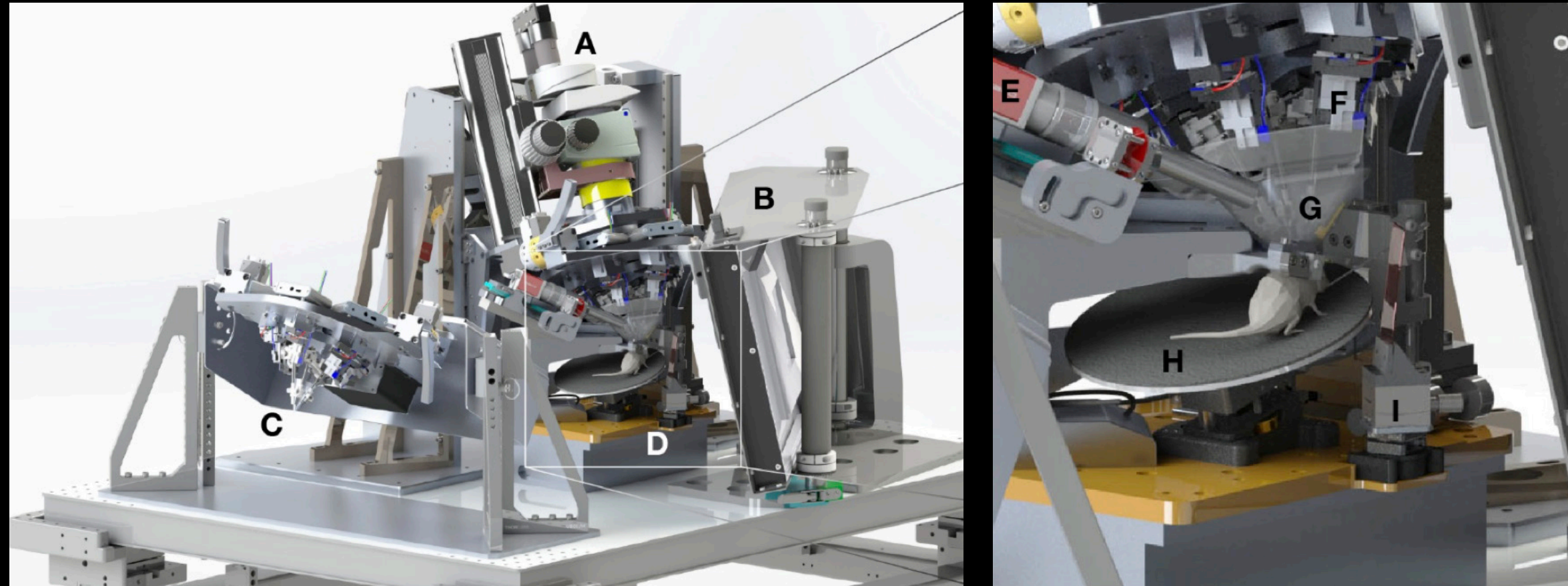


Figure 12. Recording rig.

- (A), Stereomicroscope used to visualize the insertion site. The eyepieces have been replaced with cameras, to enable remote viewing while the rig is enclosed.
- (B), Visual stimulus monitor.
- (C), Maintenance stand used to repair and replace probes, headstages, manipulators, and ground wires.
- (D), Behavior stage.
- (E), Eyetracking camera.
- (F), Manipulators used to align probes with visual areas.
- (G), Protective cone, to prevent the mouse's tail from contacting the probes.
- (H), Running wheel.
- (I), IR dichroic, to reflect the mouse's eye without obscuring the view of the stimulus monitor.

Data Structure

RAW Information of Neural Spike Data

Frame 0



Frame 100



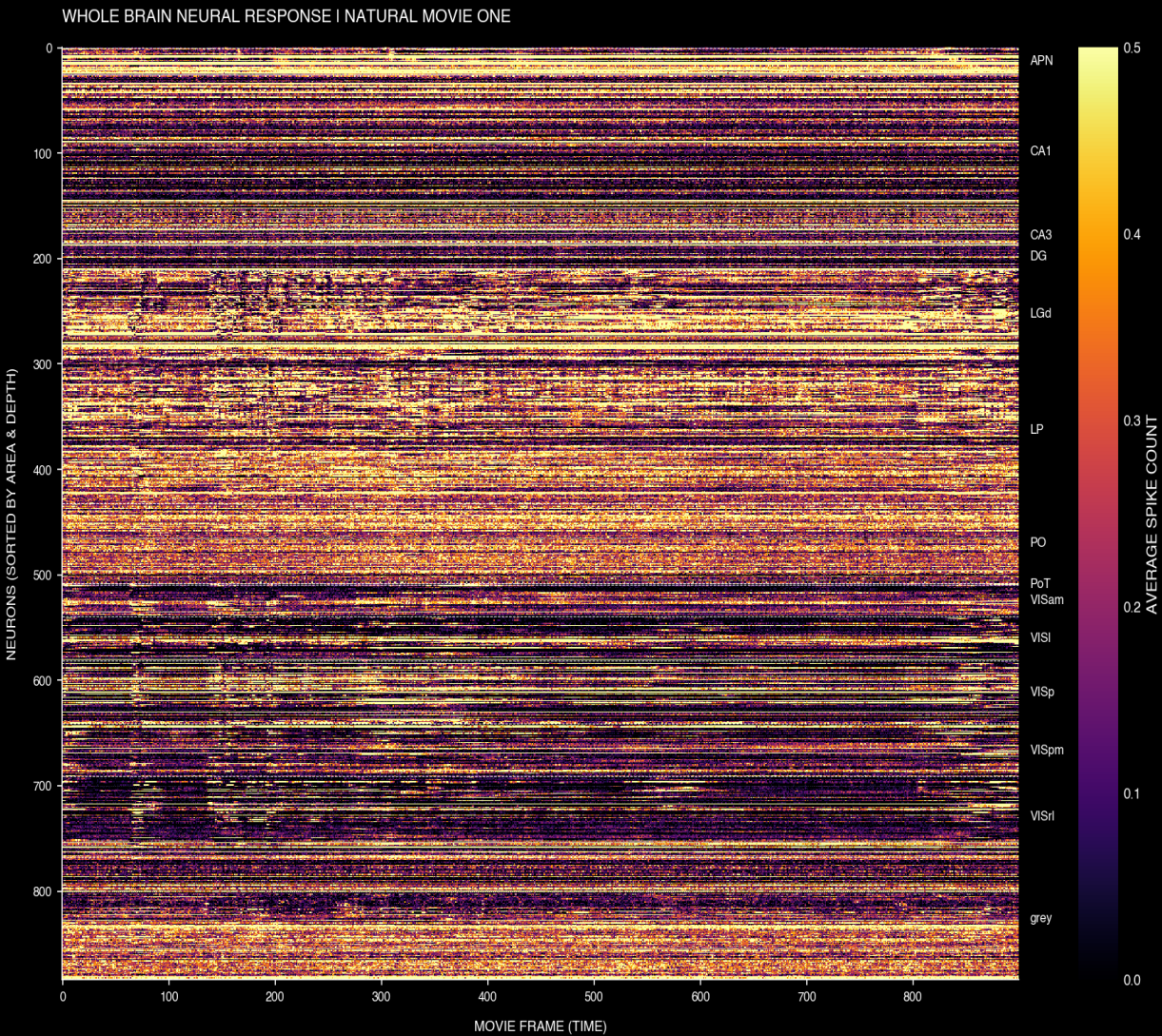
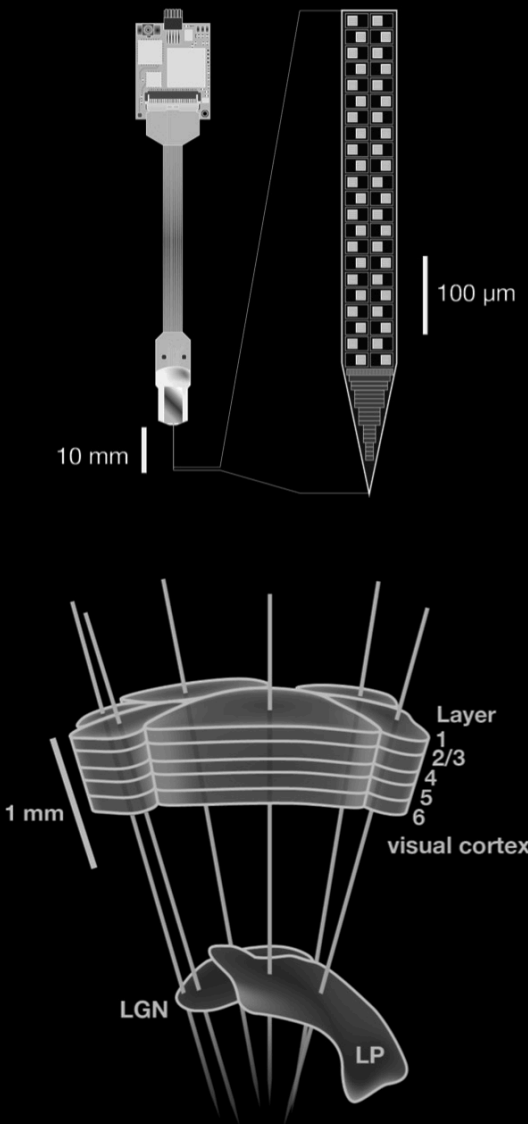
Frame 200



Frame 300



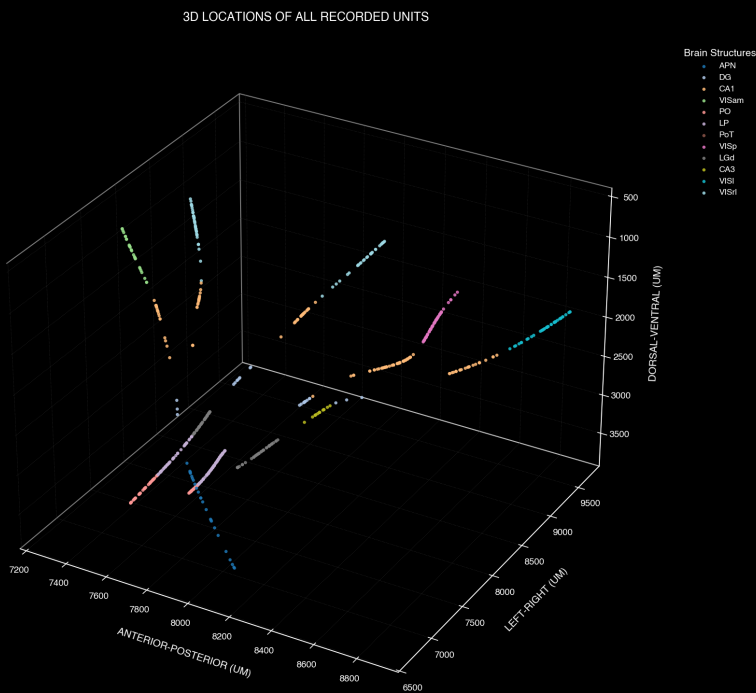
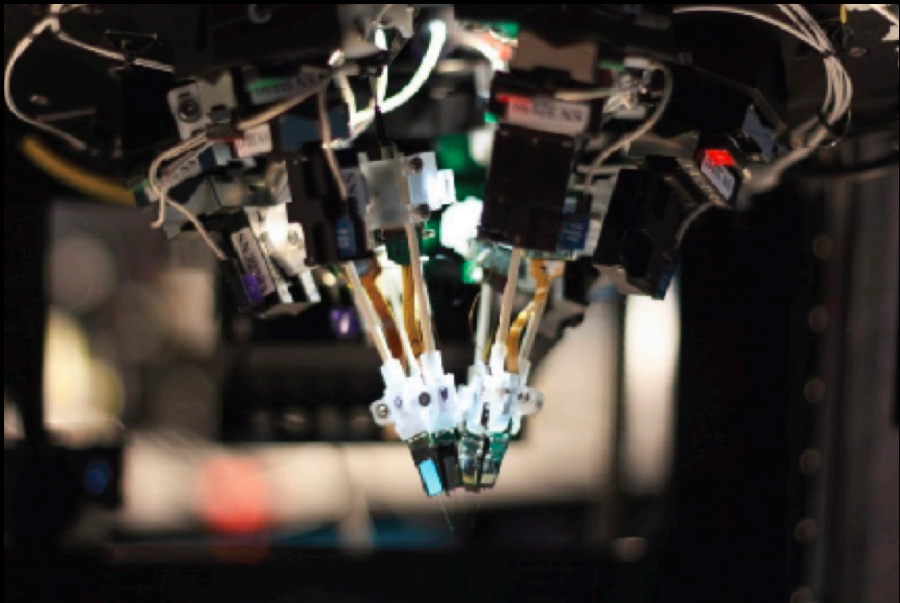
Frame 400



Overview - Target Areas
This dataset is part of an effort to understand how neurons distributed across many brain regions coordinate to produce complex behaviors (e.g., reading) by capturing the spiking activity of many neurons simultaneously. In particular, this "Visual Coding : Neuropixels" dataset is a systematic survey of spiking activity in the mouse visual system and beyond.

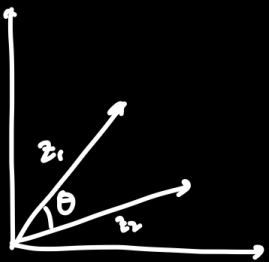
Data Structure and Stability

CCA : Stability of Latent Variable from Session and Probe



3. CCA: Canonical Correlation Analysis

- Stability of low-dim latent variable by session
- Purpose : How Y_1, Y_2 is similar?
- Approach :



$$z_1 = Y_1 u_1 \quad z_2 = Y_2 u_2$$
$$\cos \theta = \frac{z_1 \cdot z_2}{\|z_1\| \|z_2\|} = \frac{u_1^T Y_1^T Y_2 u_2}{(u_1^T \Sigma_1 u_1)^{\frac{1}{2}} (u_2^T \Sigma_2 u_2)^{\frac{1}{2}}} = \rho$$

→ $\rho_{z_1 z_2} = \rho$

$$\arg \max_{u_1, u_2} \rho(u_1, u_2) \Rightarrow L = u_1^T \Sigma_1 u_1 - \lambda_1 (u_1^T \Sigma_1 u_1) - \lambda_2 (u_2^T \Sigma_2 u_2)$$

$$\text{Q } \frac{\partial L}{\partial u_1} = \Sigma_1 u_1 - \lambda_1 \Sigma_1 u_1 = 0 \quad \rightarrow u_1^T \Sigma_1 u_1 = \lambda_1 u_1^T \Sigma_1 u_1 = \lambda_1$$

$$\text{Q } \frac{\partial L}{\partial u_2} = \Sigma_2 u_2 - \lambda_2 \Sigma_2 u_2 = 0 \quad \rightarrow u_2^T \Sigma_2 u_2 = \lambda_2 u_2^T \Sigma_2 u_2 = \lambda_2$$

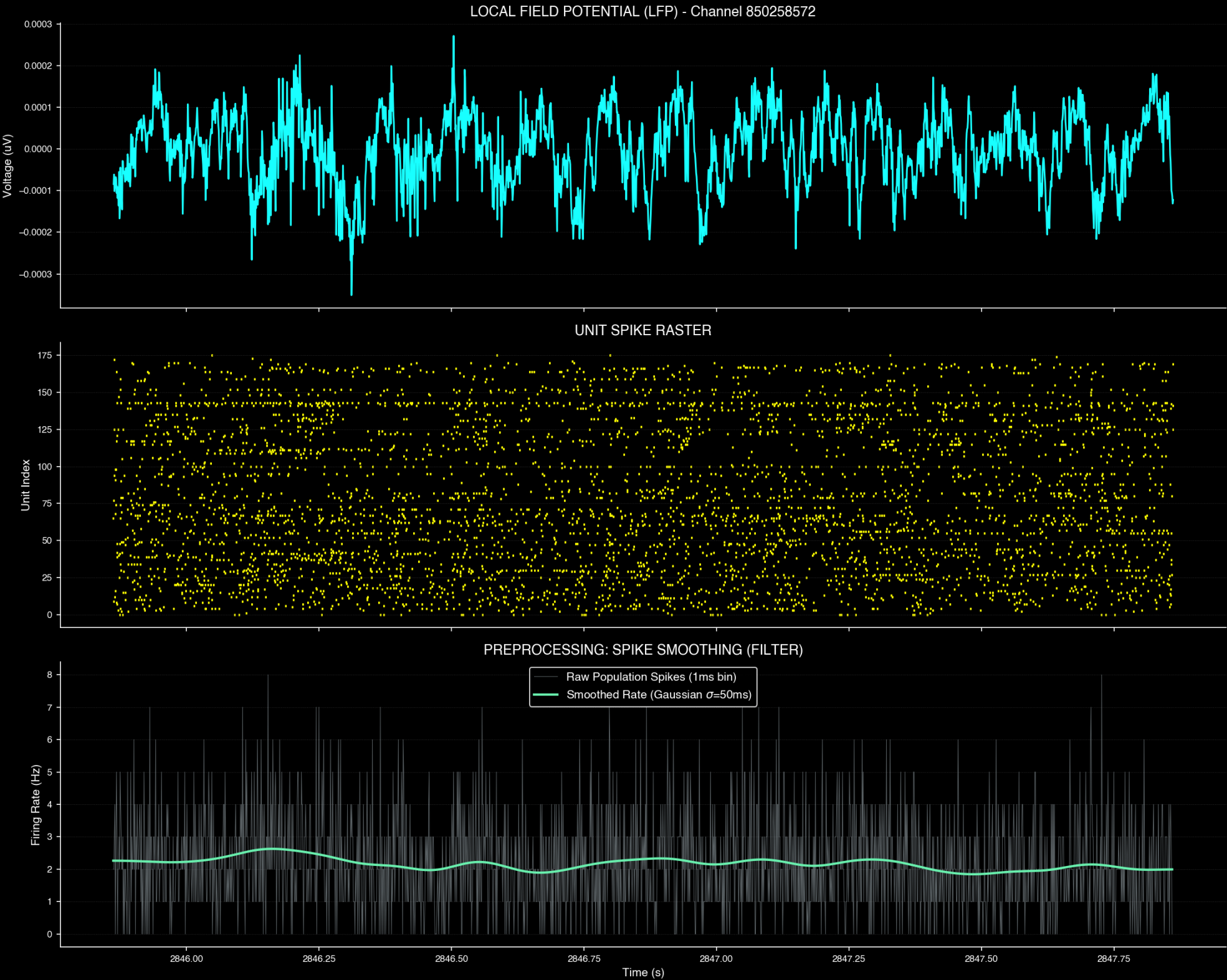
$$\rightarrow \lambda_1 = \lambda_2 := \lambda$$

$$\begin{cases} \Sigma_{11} u_1 = \lambda \Sigma_{11} u_1 \\ \Sigma_{12} u_2 = \lambda \Sigma_{12} u_2 \end{cases} \rightarrow \begin{pmatrix} 0 & \Sigma_{12} \\ \Sigma_{21} & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \lambda \begin{pmatrix} \Sigma_{11} & 0 \\ 0 & \Sigma_{22} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$



Data Structure and Stability

Spike Filter : Binary -> Continuous Data



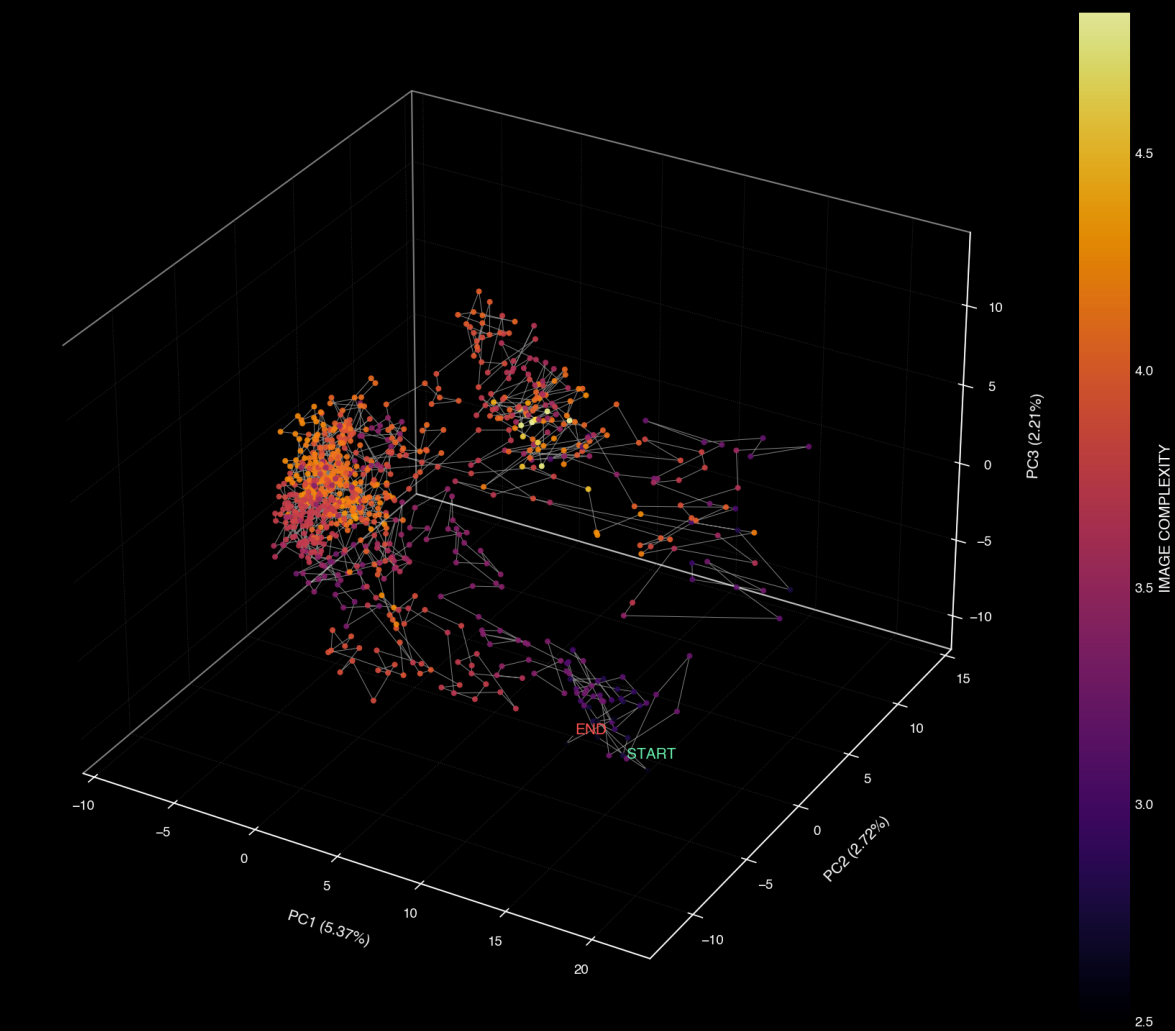
◀ Binary Spike to Continuous Variable

To obtain a continuous representation of neural population activity, each neuron's discrete spike train is first converted into a smooth firing-rate signal by convolving the binary spike sequence with a temporal filter, such as a Gaussian or exponential kernel. This filtering step removes the sparsity of the raw spike trains and captures the gradual changes in neural activation over time, producing a continuous-valued time series for every neuron.

Bring High-Dimensional Data to Lower Dimension

PCA : Principle Component Analysis - Find Linear Combination of Signal with Most Information

NEURAL POPULATION TRAJECTORY (PCA)



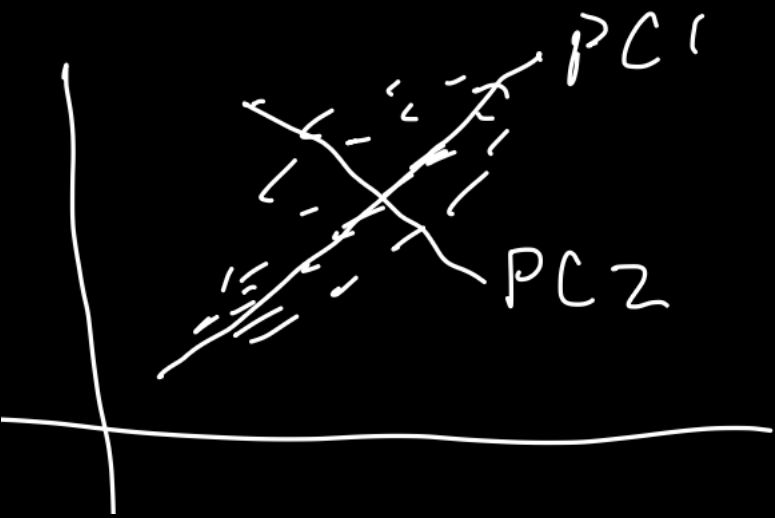
1. PCA → dimensional reduction

- high-dimensional data : most variance is in small # of dimension
- Only use the dimension w/ ↑ variance

* Covariance Matrix : $C \equiv \frac{1}{n} X^T X$

$$= \begin{pmatrix} \langle x_1, x_1 \rangle & \langle x_1, x_2 \rangle & \langle x_1, x_3 \rangle & \dots & \langle x_1, x_n \rangle \\ \langle x_2, x_1 \rangle & \langle x_2, x_2 \rangle & & & \\ \langle x_3, x_1 \rangle & & \ddots & & \\ \vdots & & & \ddots & \\ \langle x_n, x_1 \rangle & \dots & \dots & \dots & \langle x_n, x_n \rangle \end{pmatrix}$$

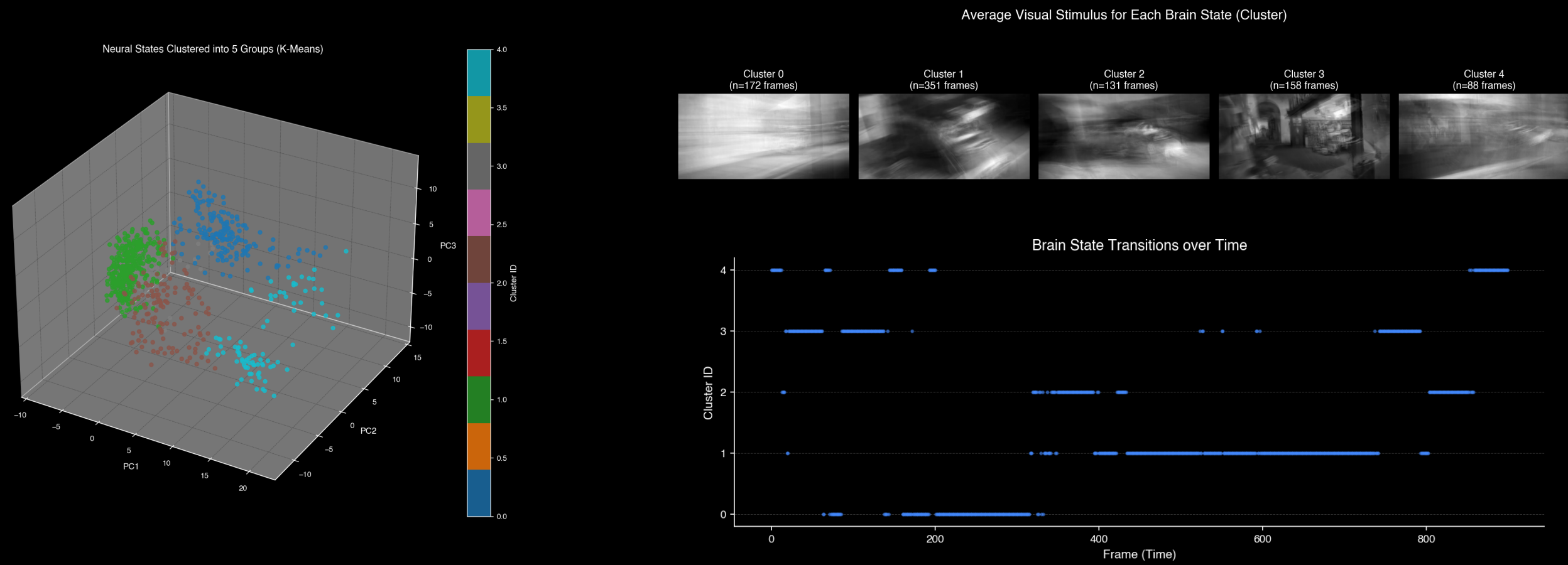
X in SVD form, let, $X = U \Sigma V^T$



◀ **PCA (Principal Component Analysis)** is a method that simplifies complex data by finding the most important patterns inside it. Imagine you have data with hundreds of variables—PCA looks for the directions in which the data changes the most and uses those directions to create a simpler version of the data.

Bring High-Dimensional Data to Lower Dimension

PCA : Principle Component Analysis - Find Linear Combination of Signal with Most Information

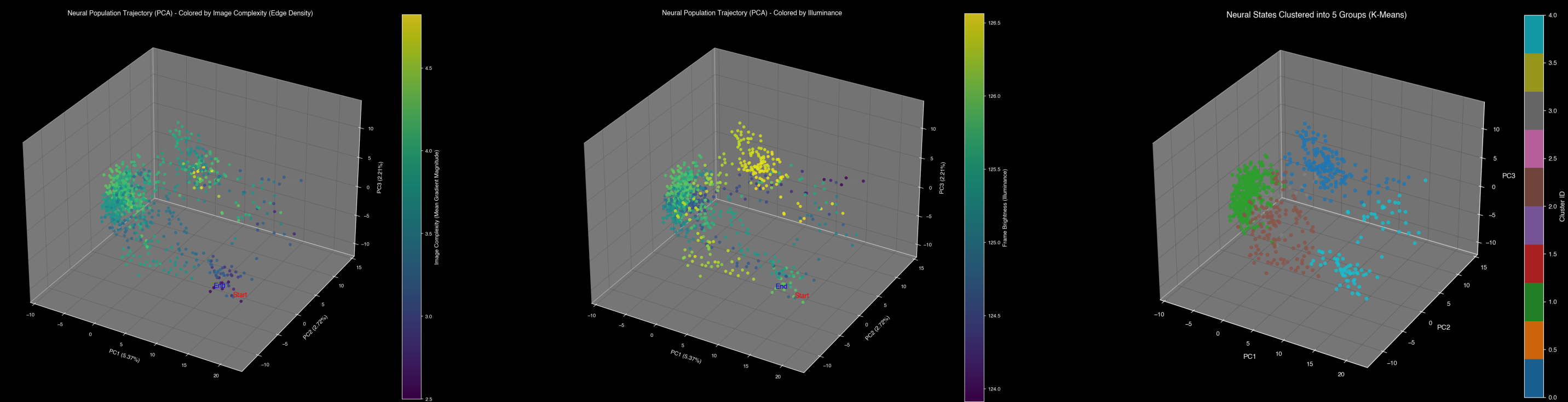


▲ K-Mean Cluster with Stimulus Average

After performing k-means clustering on the PCA-transformed neural activity, we averaged the video frames corresponding to each cluster. This produced representative stimulus segments that capture the typical visual patterns associated with distinct neural states, allowing us to compare how different portions of the urban scene relate to the organization of the neural manifold.

Bring High-Dimensional Data to Lower Dimension

PCA : Principle Component Analysis - Find Linear Combination of Signal with Most Information



▲ Luminosity Clustered

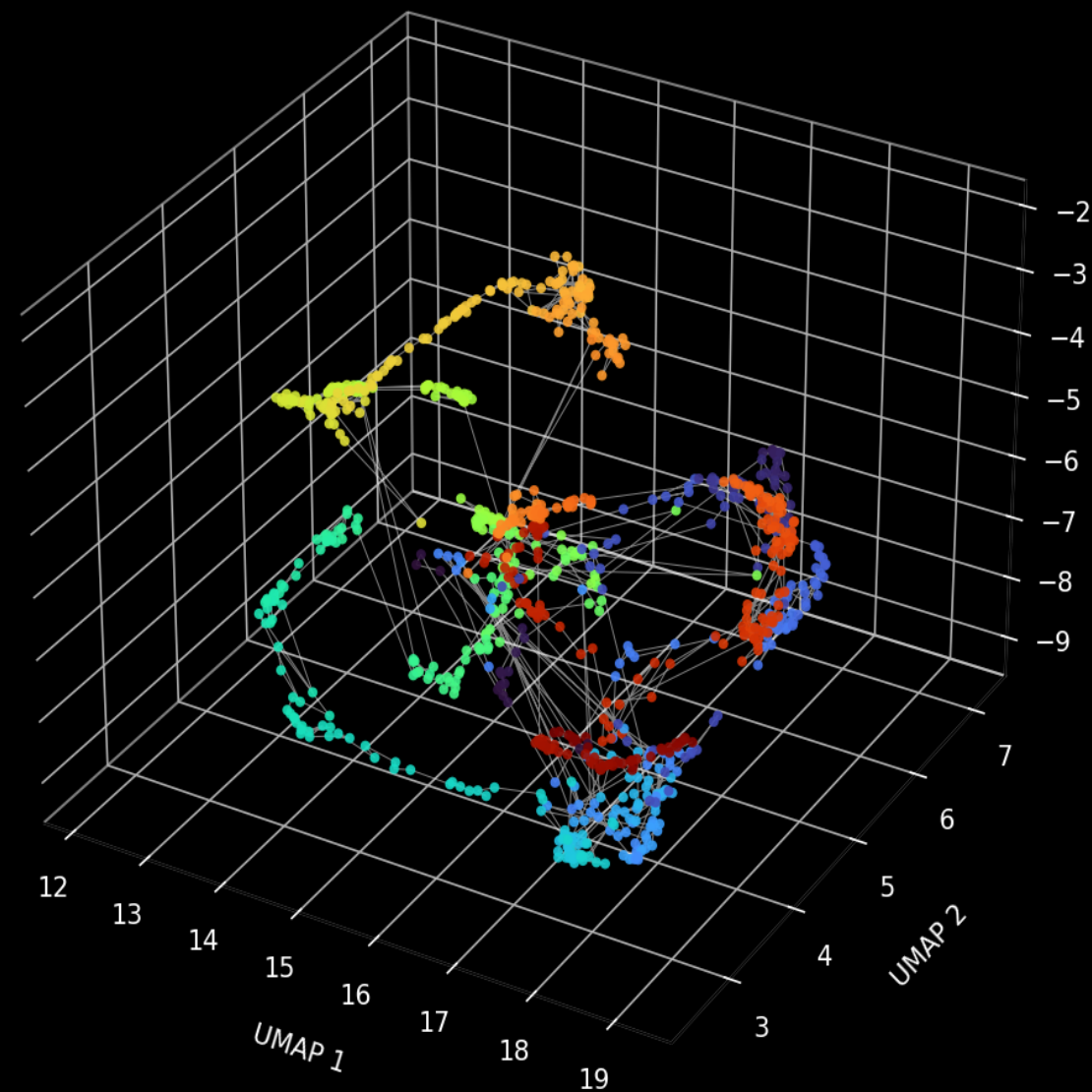
▲ Complexity Cluster

▲ K-mean Cluster

Bring High-Dimensional Data to Lower Dimension

UMAP : Uniform Manifold Apprximation and Projection for Dimension Reduction

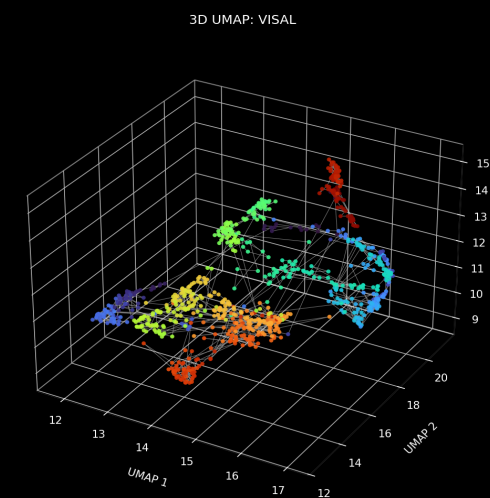
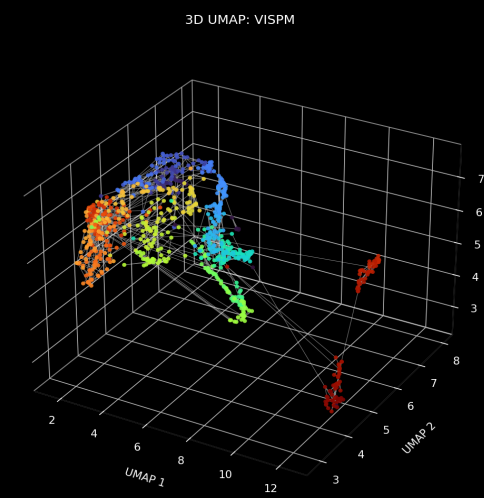
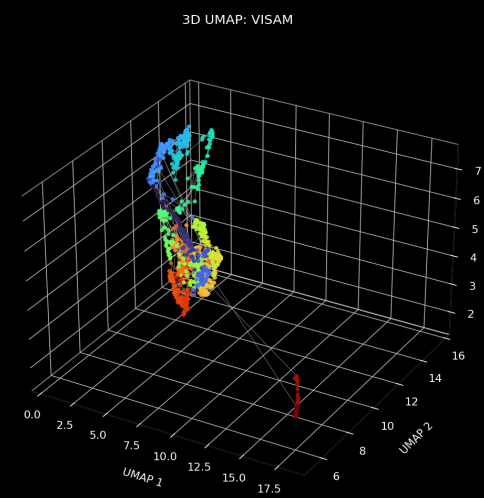
3D UMAP: VISP



High-dimensional data lie on a manifold, and the goal is to reconstruct that manifold in a lower-dimensional space.

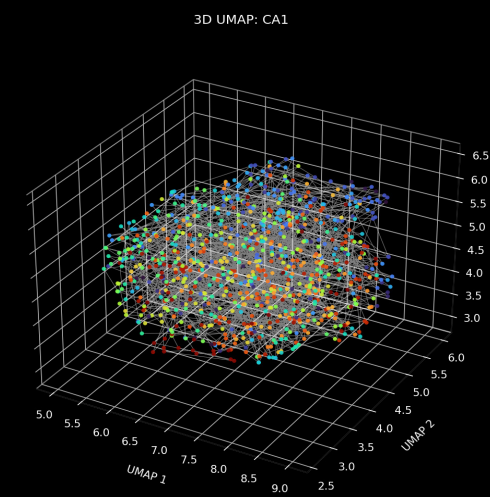
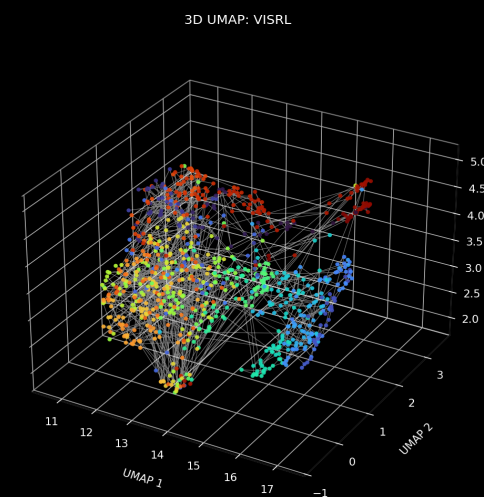
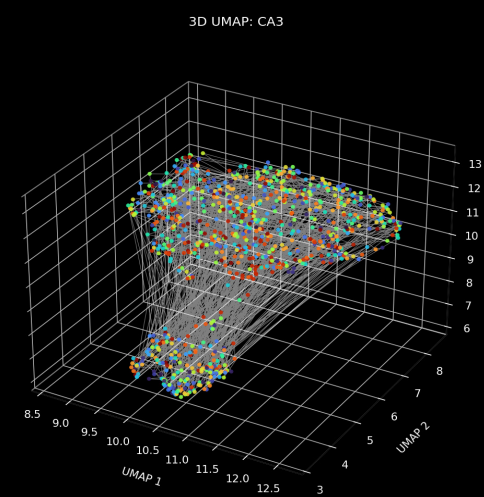
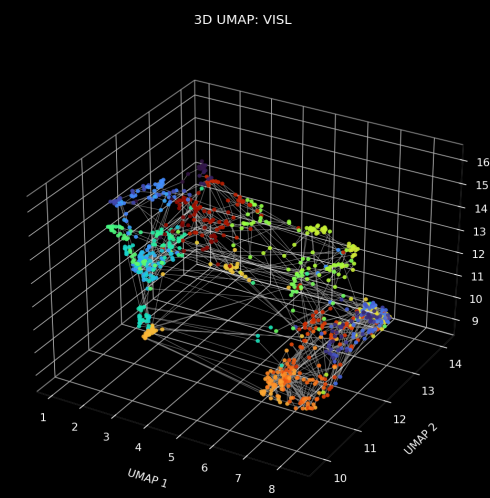
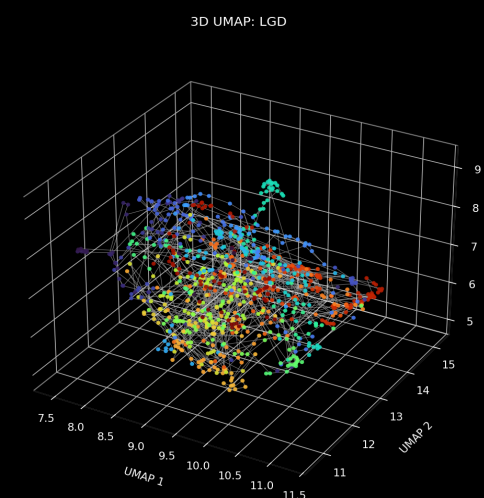
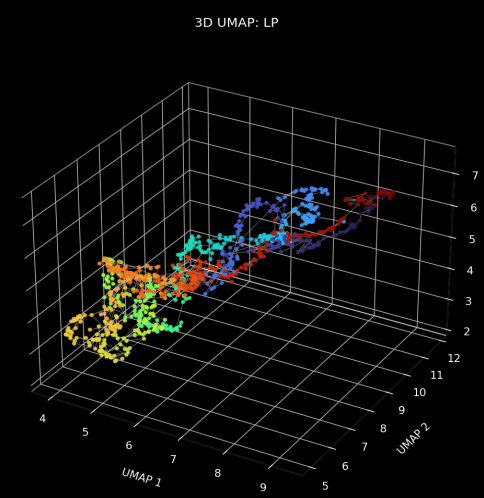
Unfolding Higher Dimension Geometry

UMAP : Uniform Manifold Appriximation and Projection for Dimension Reduction



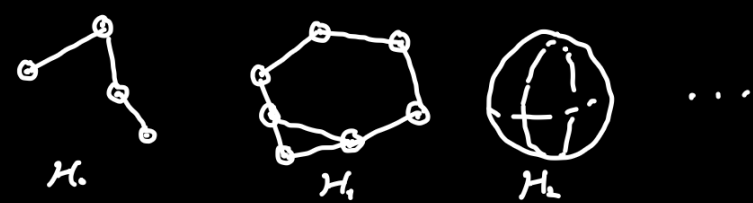
UMAP reveals the intrinsic manifold structure of the neural population activity.

Because each brain region encodes information using different circuit architectures, coding schemes, and dynamical regimes, the shape of the manifold differs from region to region.



Unfolding Higher Dimension Geometry

TDA : Barcode Analysis

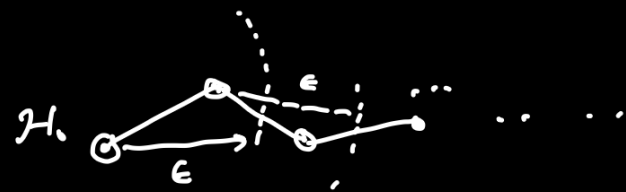


1. Filtered Simplicial Complex

$$C = \{(\sigma_i, \tau_i)\} \quad \begin{cases} \sigma_i : \text{simplex} \\ \tau_i : \text{filtration value (e.g. distance } \epsilon) \end{cases}$$

2. Persistence Homology

$$H_k^{(i,j)}(C) = \text{im} \left(\pi_k : H_k(C_i) \rightarrow H_k(C_j) \right)$$

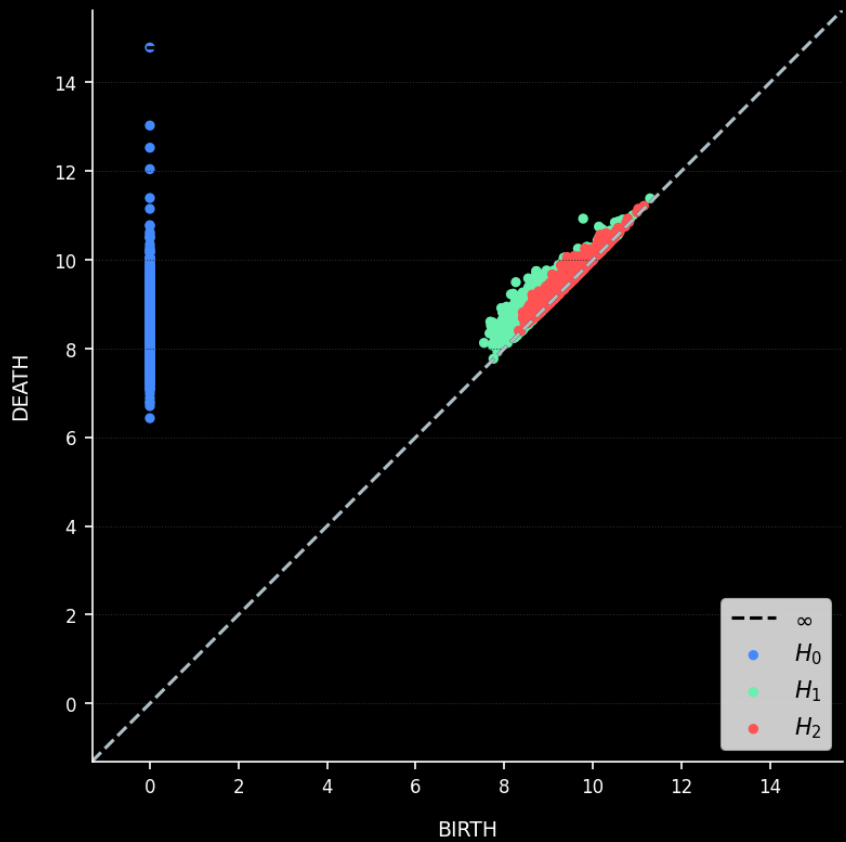


→ Homology class $[\tau, \tau_i)$ gen

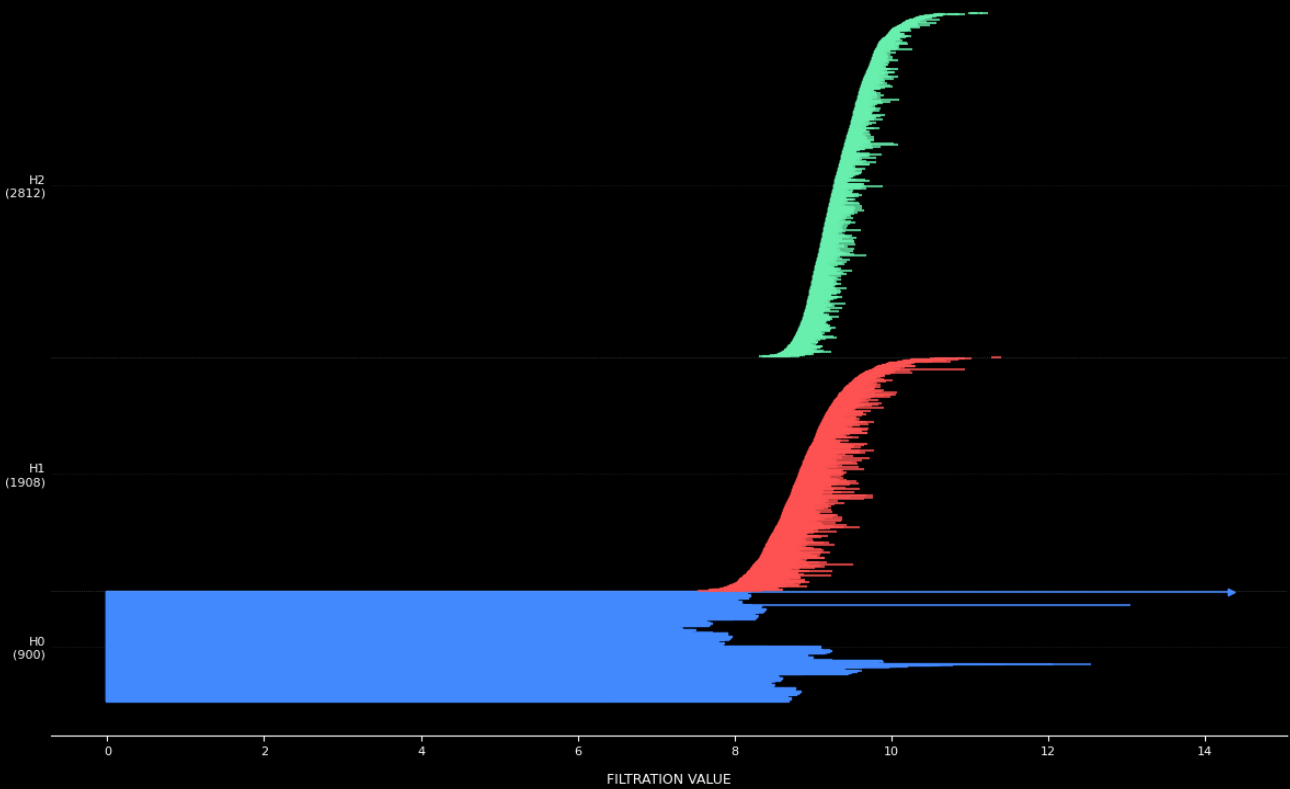
2. Structural Theorem

$$H_k(C; F) \cong \bigoplus_i \pi^{k_i} F[\pi] \oplus \left(\bigoplus_j \pi^{j_j} (F[\pi]/\pi^{j_j}) \right)$$

PERSISTENCE DIAGRAM



PERSISTENCE BARCODE



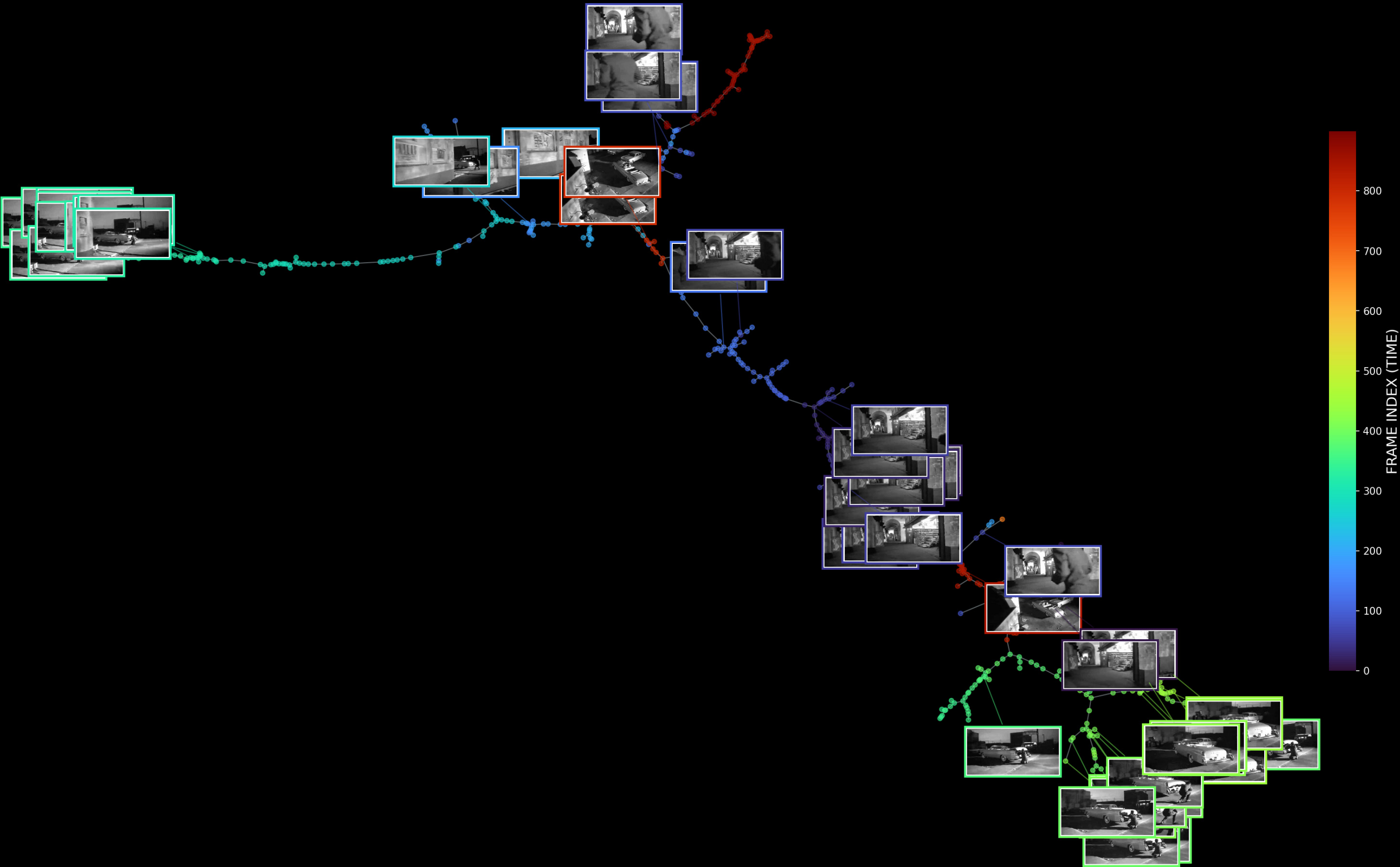
◀ Persistent Homology(PH) Analysis and Inherent Geometry

e applied persistent homology to quantify the topological structure of the neural trajectories across scales. By tracking how connected components and loops appear and disappear, PH reveals the underlying geometry of the manifold. In our analysis, the presence of long-lived H_0 features indicates that the data are organized along a largely one-dimensional, tree-like structure, where clusters merge gradually rather than forming higher-dimensional loops. This provides a topological confirmation of the linear branching patterns suggested by PCA and UMAP.

Neural Manifold Mapping

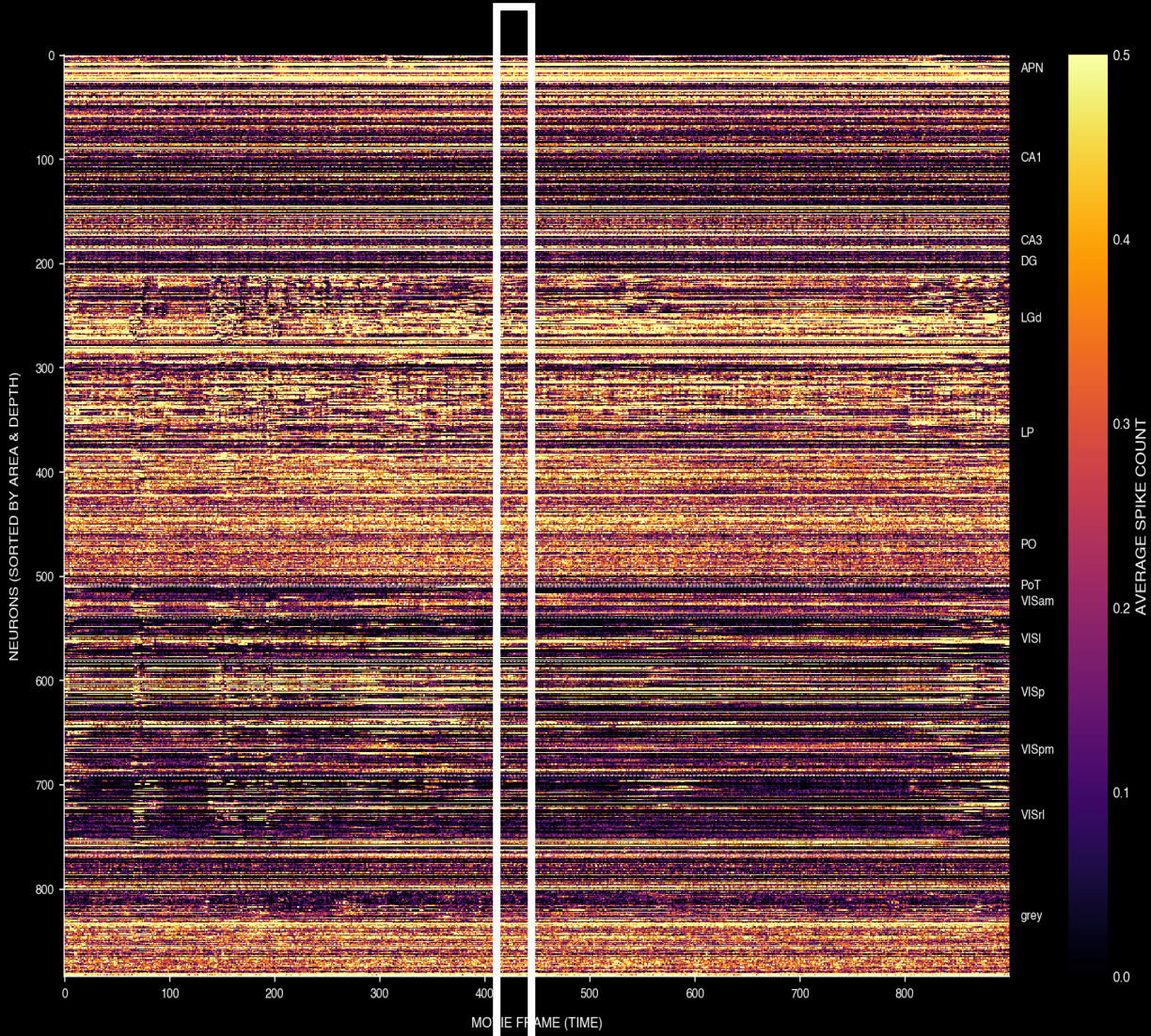
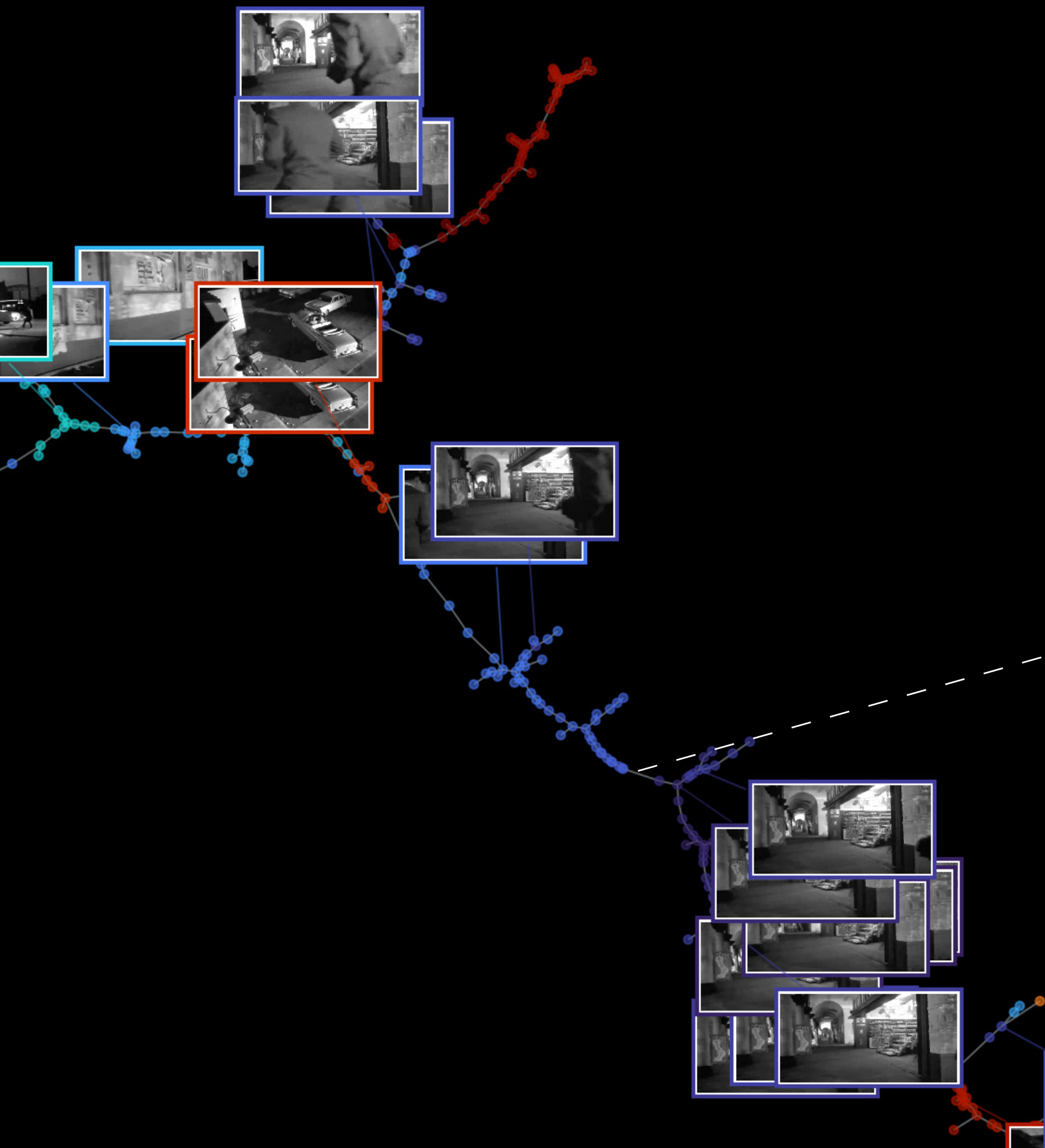
TDA : 1-Dimensional Tree

NEURAL MANIFOLD SKELETON: VISP



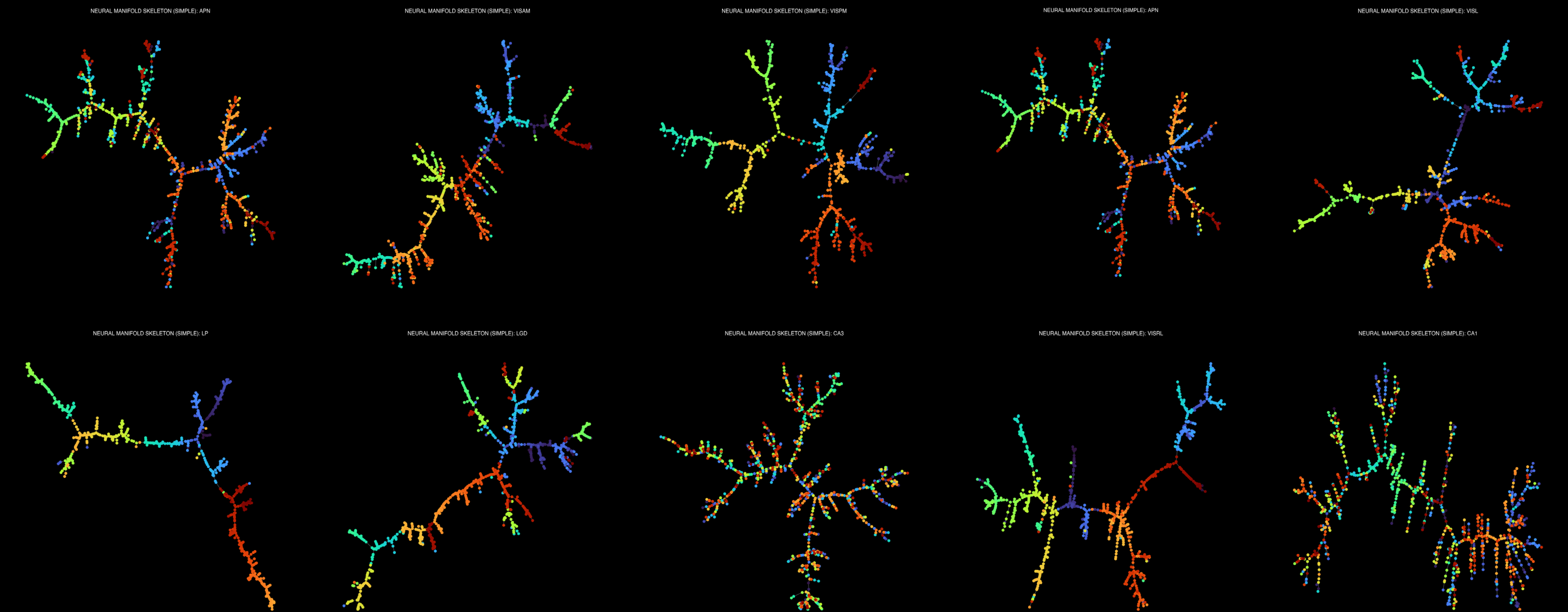
Neural Manifold Mapping Revisit

TDA : 1-Dimensional Tree



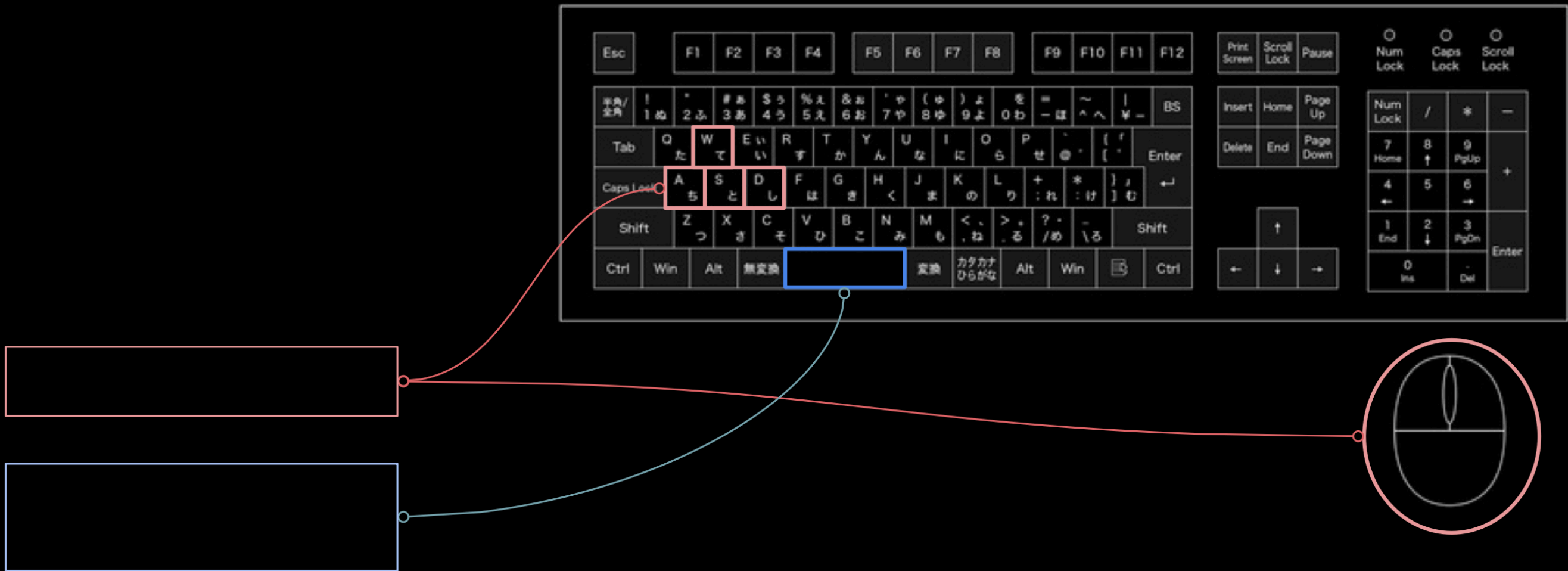
Neural Manifold Mapping

TDA : 1-Dimensional Tree

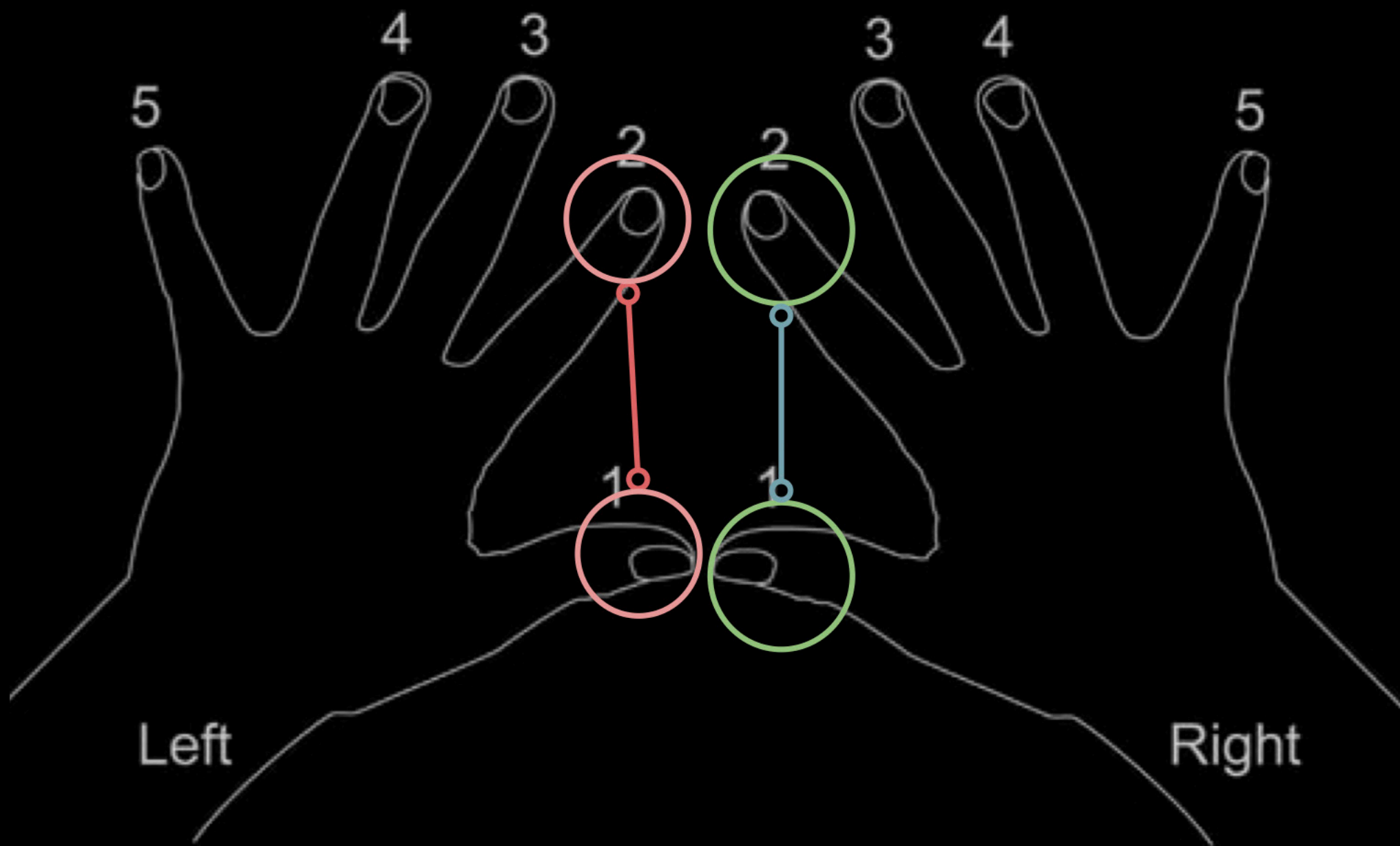


User Experience

Manipulation of High-Dimensional



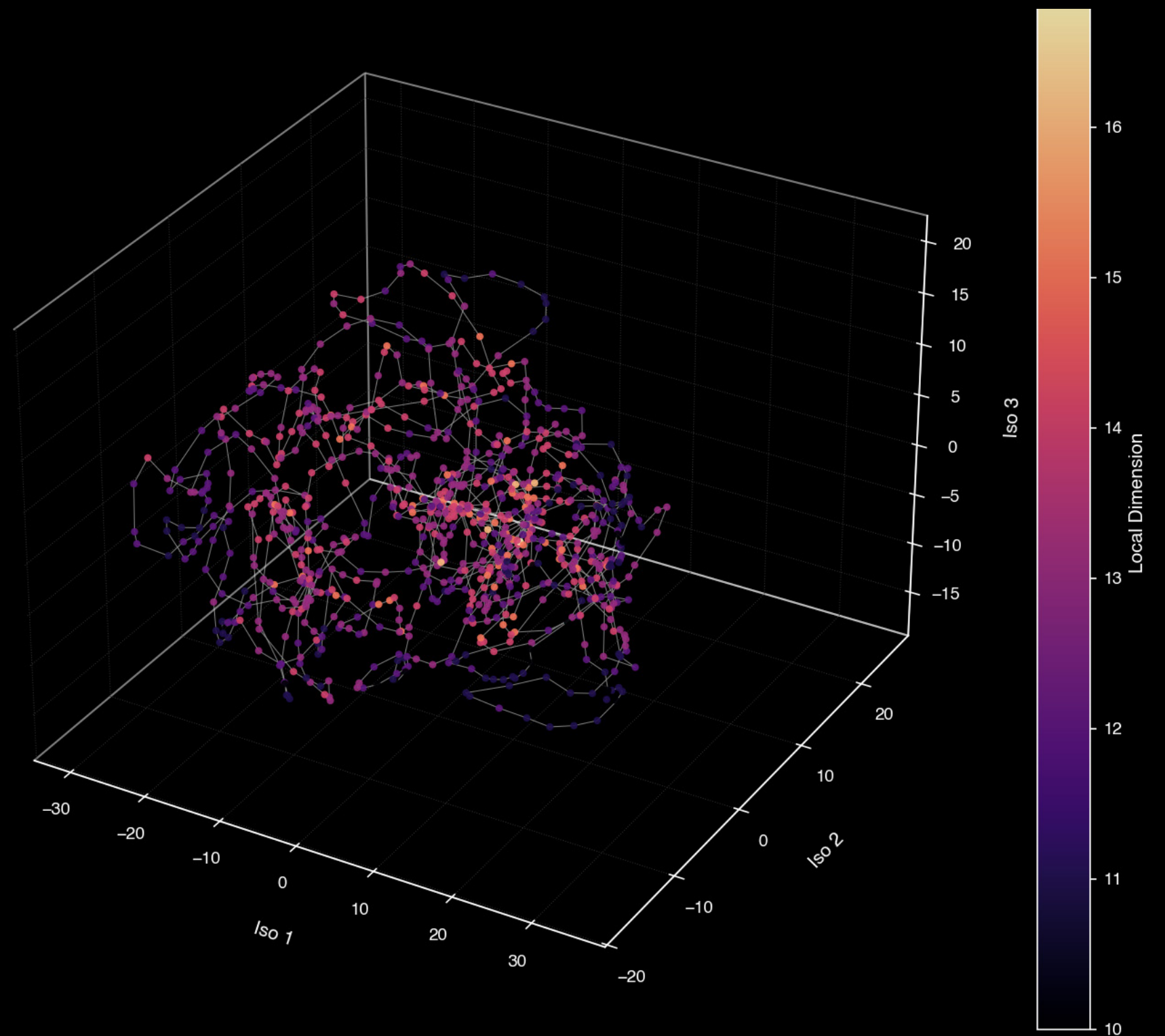
User Experience
Manipulation of High-Dimensional



Proposal : Hovering Over Dimension

High-Dimensional Data Analysis

Alternative Procedure : CA1 Topological Data Analysis



THANK YOU

